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Software note

CDMetaPOP 2: a multispecies, eco-evolutionary simulation framework for landscape demogenetics and connectivity

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Multispecies approaches to questions in ecology and evolution are at the forefront of many fields spanning both theoretical and applied research. In conservation and landscape genetics, simulation models offer a powerful means to evaluate complex questions, yet most models lack the ability to consider landscape effects on multiple species or to simulate the complex interactions that occur among species and drive eco-evolutionary processes. Tools are needed that provide a mechanistic framework to explore multiple and simultaneous interspecific interactions and their effects on demogenetic processes for each species. We present a generalized, multispecies version of Cost-Distance Meta-Population (CDMetaPOP) that simulates changes in neutral and selection-driven genotypes across space and time as a function of individual-based movement, spatial population dynamics, and multiple and dynamic landscape drivers. The number of species in a given simulation is limited only by the number of computer processors available. In addition to simulating multiple species, we use Lotka–Volterra theory to implement interspecific interactions via competition. We validate the competition model and present a real-world example to demonstrate the potential role of species interactions in landscape connectivity and gene flow. While the example uses fish as the model species, CDMetaPOP ver. 2 is flexible enough to simulate species that span the breadth of the tree of life. The simulation framework presented here provides a novel approach to addressing questions concerning the spatial ecology, evolution, and conservation of diverse taxa, from individual species to whole communities.

Keywords: competition, individual-based, landscape connectivity, landscape genetics, multispecies, simulation model, spatial population model

Introduction

Since landscape genetics connectivity emerged as a field of study 20 years ago (Manel et al. 2003), research in this area has made significant progress in using landscape patterns and processes to describe genetic structure among individuals and populations. The majority of studies in this field to date have focused on single-species



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systems in single landscapes (Storfer et al. 2010, Manel and Holderegger 2013). While highly valuable, such studies provide limited contribution to ecological theory and are limited in their applicability to the practical conservation and management of broader landscapes and species communities. Single-species applications may lead to reductive perspectives, and they falter at the nuance of eco-evolutionary questions such as the role of spatially mediated species interactions on genetic structure, or the co-evolution of communities in response to complex patterns of environmental change. Several reviews of landscape genetics have included specific calls for the development and implementation of multispecies approaches (Manel and Holderegger 2013, Balkenhol et al. 2015, Keller et al. 2015, Richardson et al. 2016), as multispecies frameworks provide a more effective tool for conservation planning and management by allowing practitioners to plan for several species or even whole communities and ecosystems.

When paired with simulation modeling, a multispecies framework can be used to explore questions that might otherwise be constrained to single species, single landscapes, or limited timeframes. Simulations that are individual-based (or agent-based) provide a powerful vehicle for exploring eco-evolutionary processes by simulating heterogeneity in individual characteristics and genotypes both within and between species (Lamarins et al. 2022). That heterogeneity can drive factors such as individual fitness or reproduction, and scales up to produce evolutionary patterns at the population level. Because of the complexity associated with eco-evolutionary multispecies models, it is essential that such simulation methods are flexible and generalizable in order to accommodate a wide range of applications.

In response to the need for generalized multispecies simulators in landscape genetics, we expanded the modeling framework Cost-Distance Meta-Population (CDMetaPOP; Landguth et al. 2017). CDMetaPOP ver. 2 can simulate populations of multiple species and their competitive interactions across complex, dynamic landscapes over time. CDMetaPOP 2 can simulate 1 to n species that occupy the same landscape, each parameterized with specific cost-distance movement processes and species-specific characteristics. Genotypes are simulated as polymorphic discretized loci passed to offspring following patterns of Mendelian inheritance, and given options for different models of mutation, natural selection, and plasticity that can be associated with environmental attributes (e.g. genotypes linked to habitat types or climate variables producing differential fitness). In multispecies simulations, species can operate independently with no interaction among them, or the user may optionally implement Lotka–Volterra competition (Lotka 1925, Volterra 1926). While species interactions occur in a variety of ways (e.g. predator–prey, parasitism), we first implemented competition not only because it is the most commonly cited interaction in the literature (Lamarins et al. 2022), but because competition is the interaction for which user demand has been the highest. The modeling framework presented here provides the opportunity to ask broad

questions about the role of functional landscape connectivity in conservation planning for communities and the effects of species interactions on eco-evolutionary processes such as gene flow, adaptation, dispersal, invasion, extinction, and range expansion.

Here we describe the basic structure and primary additions to CDMetaPOP in ver. 2, with emphasis on multispecies simulations. We then present theoretical validation of the Lotka–Volterra competition mechanism. Finally, we present a case study of competition between native westslope cutthroat trout *Onchorhynchus clarkii lewisi* (hereafter cutthroat trout) and non-native brook trout *Salvelinus fontinalis* in a tributary to the Pend Oreille River in northeastern Washington State, USA. This case study demonstrates how dynamically incorporating interspecific competition into a model of landscape genetics can significantly alter demographic and genetic (demogenetic) simulation outcomes.

Material and methods

CDMetaPOP

CDMetaPOP is an individual-based model for landscape genetics and connectivity that simulates the movement of individuals and their genes across spatially explicit resistance surfaces (Landguth et al. 2017, Mims et al. 2019). CDMetaPOP can simulate up to hundreds of thousands of individuals and their genotypes in terrestrial or aquatic systems for plants or animals whenever population demographics and dispersal behavior can be described (Menon et al. 2020, Day et al. 2021). CDMetaPOP 2 maintains all functionality from the previous version, including aspects such as temporally dynamic landscapes, neutral and adaptive selection, disease transmission, and intra-specific competition, among others (Fig. 1). Landscapes in CDMetaPOP are divided into ‘patches’ that house subpopulations and provide a common environment for occupying individuals. Patches govern processes such as carrying capacity, habitat suitability, and environmental selection. Within patches, individuals are explicitly simulated and characteristics are tracked such as growth, maturation, movement, fitness, and genotype. Movement among patches (e.g. dispersal, mating, migration) is based on a weighted random draw from a cost distance matrix that is most commonly based on a landscape resistance surface. Most processes in CDMetaPOP are spatially and/or temporally dynamic to implement effects of spatial heterogeneity and to accommodate changing environments or updated management plans over time.

Primary additions in CDMetaPOP 2 include parallel processing (i.e. utilization of multiple computer processors) to accommodate simultaneous simulation of multiple species across the same landscape, implementation of competition among species integrated into age-structured density-dependent mortality, and an update from Python 2.7 to Python 3.8. Figure 1 provides a step-by-step conceptual work flow of CDMetaPOP 2.

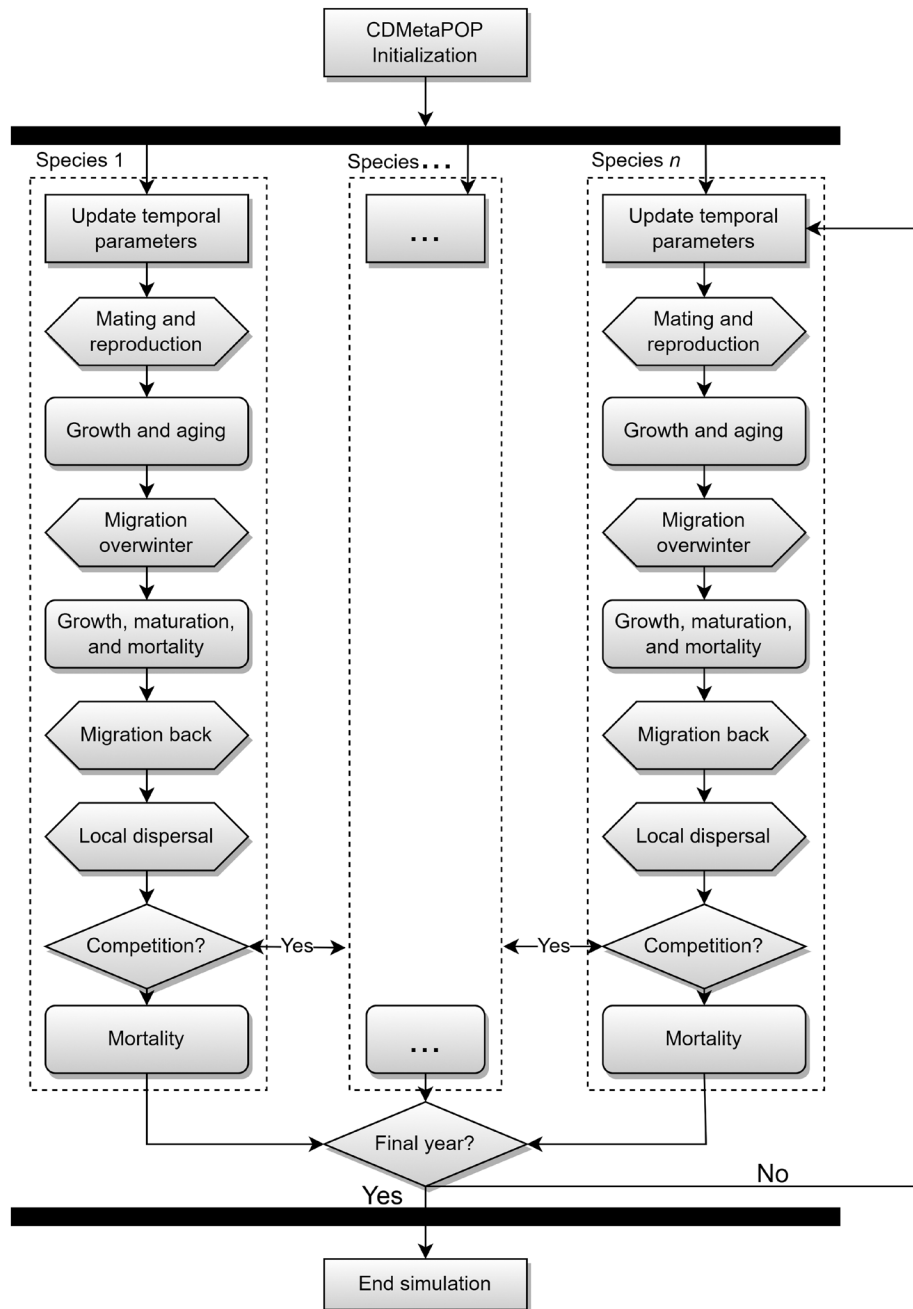


Figure 1. CDMetaPOP 2 processes, timing, and parallelization. CDMetaPOP 2 is initialized by loading input parameters and then creating queue data structures in which to house data to be communicated among processes for multispecies applications. A number of instances of CDMetaPOP 2 equal to the number of species in the simulation are then assigned to individual processors where they run independently until the competition step. Each year, landscapes and model parameters can be updated to reflect imposed changes in the environment or in the population over time. Mating involves searching for a mate based on a user-defined probability function followed by production of offspring whose alleles are assigned according to Mendelian inheritance. Growth can be deterministic based on an age–size relationship or can be a function of the environment (e.g. temperature). Migration is optional and movement is based on a probability draw from a cost surface. Between migration movements, growth may occur again along with maturation processes and mortality. Probability of mortality is the product of several possible sources including density-dependence, age-based survival, selection, and independent processes available to simulate management actions or stochastic environmental effects. Following migration back to natal grounds, local dispersal may be implemented following the same process as migration movement. After dispersal, the software checks to see if competition will be implemented among species and pauses until all species have communicated their demographic status to the other processes. The subsequent mortality module repeats the mortality previously described, but now accounts for Lotka–Volterra competition. The simulation then cycles back to implement any updates to temporal parameters and this continues until the user-defined number of simulation years is reached.

Parallel processing allows for efficient use of increasingly ubiquitous multi-core processors. Simulations of multiple, non-interacting species are straightforward to run in parallel, as all mechanisms for each species are considered independent (i.e. no communication among child processes is needed). We implemented parallel processing in CDMetaPOP 2 using the 'multiprocessing' library from Python ver. 3.8. As each species occupies one processor, the number of species in a simulation is limited to the number of available processors. These processes run independently and in parallel until the density-dependent mortality module is called (Fig. 1). At this point, each species' process communicates the abundance, spatial distribution, and size/age distribution of their respective populations to all other species' processes using the Python 'queue' data structure. This information is incorporated into Leslie matrix projections modified to match logistic population growth (Miller and Ankley 2004) and Lotka–Volterra competition.

Competition mechanism

Density-dependent mortality in CDMetaPOP occurs at the level of the patch rather than the individual, although individual-level traits such as fitness and genotype can affect probability of mortality. Interspecific competition and its associated mortality are likewise patch-based, and the strength of competition among species can vary spatially with the environment. Competition is based on the Lotka–Volterra family of equations (Lotka 1925, Volterra 1926, Wangersky 1978) which take the form (Eq. 1 and 2):

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{K_1 - (N_1 + \alpha_{21} N_2)}{K_1} \right) \quad (1)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{K_2 - (N_2 + \alpha_{12} N_1)}{K_2} \right) \quad (2)$$

where N represents the population size of the respective species, K represents carrying capacity, r represents the intrinsic rate of population growth, and α represents the competition coefficient, which can be asymmetrical between species. These equations differ from the classic logistic growth function by adjusting the proportion of K occupied as a function of the product of N and α of the competitor species. To implement this effect, we combined Lotka–Volterra competition with an age-structured, density-dependent logistic growth model described by Miller and Ankley (2004) and applied independently within each patch (Eq. 3):

$$n_{t+1,j} = \exp(-rP_{t,j} / K_{t,j}) M_{t,j} n_{t,j} \quad (3)$$

where n (subscript t, j) or $n_{t,j}$ is the vector of population age structure at time t in patch j , M is the Leslie matrix (Leslie 1945), and $P_{t,j}$ is population size at patch j . Because CDMetaPOP is individual-based and not able to

accommodate fractions of individuals from the Leslie model projections, we incorporated a stochastic process to round the projected number of survivors per age class up or down as a function of a random draw on the remainder. This adjustment made the Leslie matrix model more robust to simulations with low population sizes or skewed age structures.

To implement the Lotka–Volterra competition effect, we adjusted the ratio of $P_{t,j}/K_{t,j}$ to account for each competitor species and their patch-specific competition coefficient when implementing density-dependent mortality, so that (Eq. 4):

$$P'_{t,j} = P_{t,j} + \sum_2^n P_{x,t,j} \alpha_{x1,t,j} \quad (4)$$

where x corresponds to each species and n is the number of species. Parameters required to implement this growth and competition model include age-structured fecundity and survival, patch carrying capacity, and competition coefficients for each species pair. Intrinsic population growth rate is derived from the Leslie matrix.

Because the strength of competitive interactions can be mediated by environmental factors (Taniguchi and Nakano 2000, Minchinton and Bertness 2003), competition coefficients may vary both spatially and temporally. This is accomplished by including α as a patch-level variable, allowing interactions among each pair of competitors to vary with the environment over space and time (e.g. due to elevation, temperature, prey availability).

Theoretical validation

We conducted several two-species simulation experiments to evaluate how well our implementation of Lotka–Volterra competition matched theoretical predictions. Initial validation scenarios were simplified to exclude spatial processes by simulating a single patch, testing the following three cases across a range of starting conditions: 1) competitive exclusion in which the same species always wins, 2) stable coexistence between species, and 3) competitive exclusion in which either species may win. These are generalized simulations in which both species were assigned the same parameters but varied by competition coefficient α , carrying capacity K , and starting population size P . Leslie matrices and other simulation parameters were based on previously published models of brook trout (Schill et al. 2017, Day et al. 2018). Conditions for each test case were selected to include all possible conditions present in the isocline graph for each of the three scenarios (Table 1).

Empirical application

To demonstrate the effects of landscape configuration and interspecific competition on genetic diversity and population structure, we developed a case study grounded in a real-world system where competition occurs between native cutthroat trout and non-native brook trout (Fig. 2). This system is

Table 1. Parameter values used for theoretical validation of test cases for Lotka–Volterra competition. K is carrying capacity, P is starting population size, and α_{ij} is the competition coefficient of species i affecting species j .

Scenario	Case	K_1	K_2	P_1	P_2	α_{21}	α_{12}
1	1	1000	600	200	200	0.8	1.0
2	1	1000	600	400	400	0.8	1.0
3	1	1000	600	700	700	0.8	1.0
4	1	1000	600	200	200	0.9	0.6
5	1	1000	600	400	400	0.9	0.6
6	1	1000	600	700	700	0.9	0.6
7	2	1000	500	400	200	1.0	0.3
8	2	1000	500	1200	150	1.0	0.3
9	2	1000	500	150	650	1.0	0.3
10	2	1000	500	1200	600	1.0	0.3
11	3	1000	1000	300	100	2.0	2.0
12	3	1000	1000	100	300	2.0	2.0
13	3	1000	1000	800	300	2.0	2.0
14	3	1000	1000	300	800	2.0	2.0
15	3	1000	1000	500	150	2.0	2.0
16	3	1000	1000	150	500	2.0	2.0

located in Sullivan Creek, a tributary to the Pend Oreille River in northeastern Washington State, USA, and is representative of many stream systems in the western United States where historic brook trout introductions have contributed to the decline and displacement of native cutthroat trout through competition and predation (Dunham et al. 2002). Cutthroat trout in the system follow a commonly observed pattern of higher abundance in the headwater and tributary reaches while brook trout tend to be more abundant in the mainstem reaches (Dunham et al. 2002). The dendritic stream networks inhabited by native fish such as cutthroat trout often contain

artificial (e.g. road crossing structures) and natural (i.e. geologic) barriers that disrupt connectivity and isolate populations from adjacent stream reaches (Shepard et al. 2005). This isolation exposes populations to genetic drift and loss of genetic diversity in addition to greater risk of extirpation from stochastic environmental and demographic processes (Peterson et al. 2014, Carim et al. 2016). Because the two species have similar life histories and to maintain a controlled simulation environment free from confounding factors due to differences between species, we parameterized the two species identically except for α (competition coefficient), which differed by species but was constant across space.

The simulation area (Fig. 2) comprised 36 stream reaches, or patches, each approximately 200 m in length. We split this region into three 12-patch stream sections representing two tributaries separated by a mainstem reach of Sullivan Creek. We simulated two isolated cutthroat trout populations in the two tributaries to Sullivan Creek with a barrier to both upstream and downstream movement located at the confluence of each tributary with the mainstem (red x's in Fig. 2). One of these populations was assigned high genetic diversity, while the other was assigned low genetic diversity. We populated the mainstem with non-native brook trout only. Following a burn-in period of 50 years, we removed the barriers to movement and ran the simulation for 500 years, allowing each species to move freely according to an empirically derived dispersal distribution, thereby colonizing previously unoccupied patches. To demonstrate the effects of competition with brook trout on landscape connectivity between cutthroat trout populations, we systematically varied the magnitude of the cutthroat trout competition coefficient

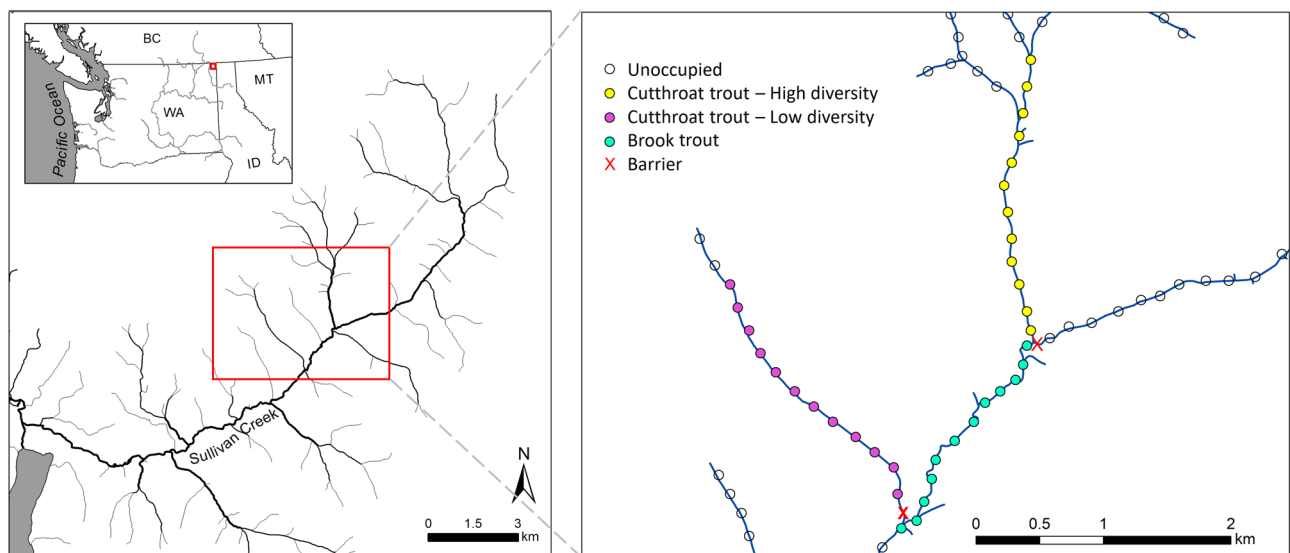


Figure 2. Study area and landscape configuration for the empirical application demonstrating the effects of competition on landscape connectivity in Sullivan Creek, a tributary to the Pend Oreille River in Washington State, USA. The landscape is divided into three stream sections: two tributaries with isolated populations of native westslope cutthroat trout separated by a mainstem reach occupied by non-native brook trout. The upstream and downstream tributaries are occupied by westslope cutthroat trout with low and high genetic diversity, respectively. The simulation is allowed a burn-in period of 50 years, at which point the barriers are removed and dispersal among the three stream sections is allowed to occur for both species. All other patches are considered unoccupied and unsuitable for purposes of the simulation.

(α_{21} i.e. the effect of brook trout on cutthroat trout) while holding the brook trout coefficient (α_{12}) constant.

Age-structured survival and fecundity for this model were derived from a long-term brook trout study (McFadden et al. 1967) following Schill et al. (2017; Supporting information), and the intrinsic population growth rate (r) derived from the associated Leslie matrix = 0.035. Carrying capacity (K) was set to 200 for all patches for each species, approximating the number of total salmonids found in a typical patch in the study system after accounting for competition. Annual dispersal distance was assigned based on a probabilistic draw on a Pareto distribution fit to observed brook trout dispersal distances (Day et al. 2020a). Because we were only interested in the genetic diversity of the native species, we only assigned and tracked genotypes of cutthroat trout. We simulated 30 microsatellite loci with 30 alleles each and assigned the low-heterozygosity tributary the minimum possible heterozygosity (i.e. all loci fixed) and the high-heterozygosity tributary the maximum possible heterozygosity (i.e. $H_o = 1$).

We varied α with values of 0, 0.2, 0.4, 0.6, 0.8, and 1, with the default value for the effect of cutthroat trout on brook trout set at 0.5. We ran 10 Monte Carlo replicates for

each competition scenario and recorded annual mean values for expected and observed heterozygosity and 25 year values for pairwise Nei's genetic distance (Nei 1972) and global Jost's D_{est} (Jost 2008).

Results

Theoretical validation

Here we present results from 50 replicates of one scenario for each test case run for 200 years (Fig. 3; see Supporting information for additional scenarios). Case 1 results in competitive exclusion, in which Species 1 always wins regardless of starting population size. Case 2 results in stable coexistence between species where the two isoclines cross. Case 3 results in competitive exclusion where either species may win depending on the starting population sizes.

Simulations for the empirical case study demonstrated that as α increased, gene flow across the mainstem decreased, as represented by change in heterozygosity, pairwise Nei's genetic distance, and global population differentiation (D_{est} ; Fig. 4).

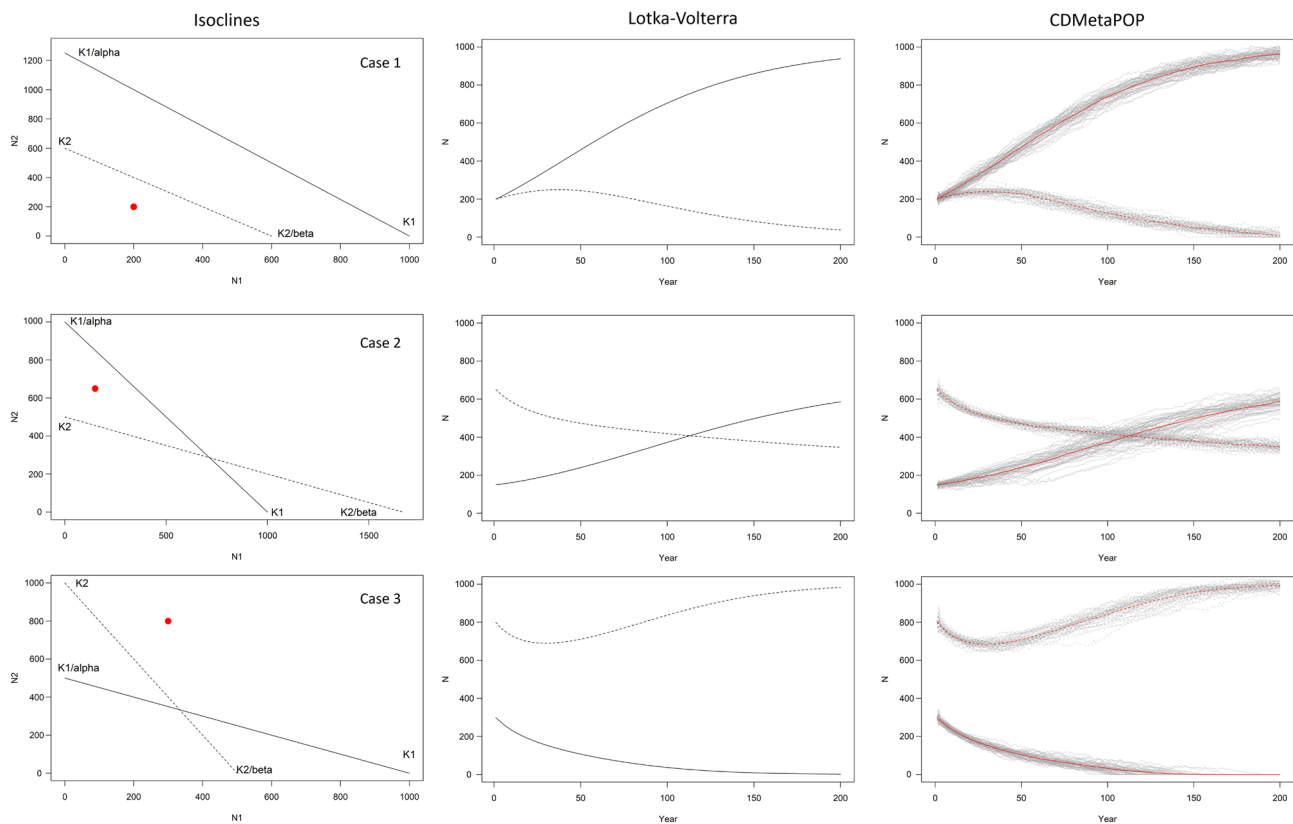


Figure 3. The left column displays isocline graphs illustrating the Lotka–Volterra competition relationship between species for each of the cases tested in CDMetaPOP 2. The solid line represents the isocline of species 1, and the dotted line represents the isocline of species 2. Competition coefficients of the effects of species 2 on species 1 and species 1 on species 2 are represented by α and β , respectively. The red dot indicates the starting conditions for population size for each species. The middle column shows population size over time based on Lotka–Volterra equations (left panel) and the right column shows 50 replicates of CDMetaPOP simulations using an age-structured, density-dependent Leslie matrix model. Case 1 results in competitive exclusion in which species 1 always wins. Case 2 results in stable coexistence between species. Case 3 results in competitive exclusion where either species may win, depending on the starting conditions.

These effects are likely due to reduced population sizes of cutthroat trout as well as difficulty in colonizing patches unoccupied by cutthroat trout as the strength of competition increases (Supporting information). In addition, a patch's position on the landscape relative to the mainstem influenced genetic diversity and genetic structure between populations, as more distal patches were less connected to the opposite cutthroat trout population than more proximal patches. This effect persisted even as genetic metrics stabilized over time (Fig. 4).

Discussion

CDMetaPOP ver. 2 contributes two primary advances to the fields of landscape connectivity and landscape genetics. First, it provides a mechanistic simulation framework to examine landscapes from the perspective of communities or guilds rather than single species. Because of its flexible design, it can accommodate multispecies simulations across a wide range of study systems, from theoretical to empirical and from fish and wildlife populations to forest stands. Second, CDMetaPOP 2 now provides the opportunity to consider landscape connectivity in the context of dynamic species interactions. Previous work has integrated factors like

interspecific competition or predation risk into models of landscape resistance (Wilkinson et al. 2013, Fletcher et al. 2019), but these are usually static implementations as species do not respond dynamically to the presence, abundance, or movement of other species. In our model, population dynamics and genetic patterns emerge from interactions between species in combination with the stochastic, individual-based processes (Grimm and Railsback 2005).

The incorporation of multispecies interactions within individual-based, eco-evolutionary simulation frameworks is not novel. Most previous applications have been system-specific or focused on macroevolution (Lamarins et al. 2022), whereas CDMetaPOP's strength lies in generalizability across species and systems with a strong focus on ecological and behavioral processes. Slim 4 (Haller and Messer 2023) is a generalizable model that also recently added multispecies interactions to its eco-evolutionary simulation framework. Such parallel development of new functionality is needed because it promotes intermodel comparison of simulation outcomes and flexibility for the user to choose the software that best fits the needs of the research question (Araya-Donoso et al. 2022).

Several existing functions from the original version of CDMetaPOP can now be combined with the multispecies functionality to explore novel eco-evolutionary questions in

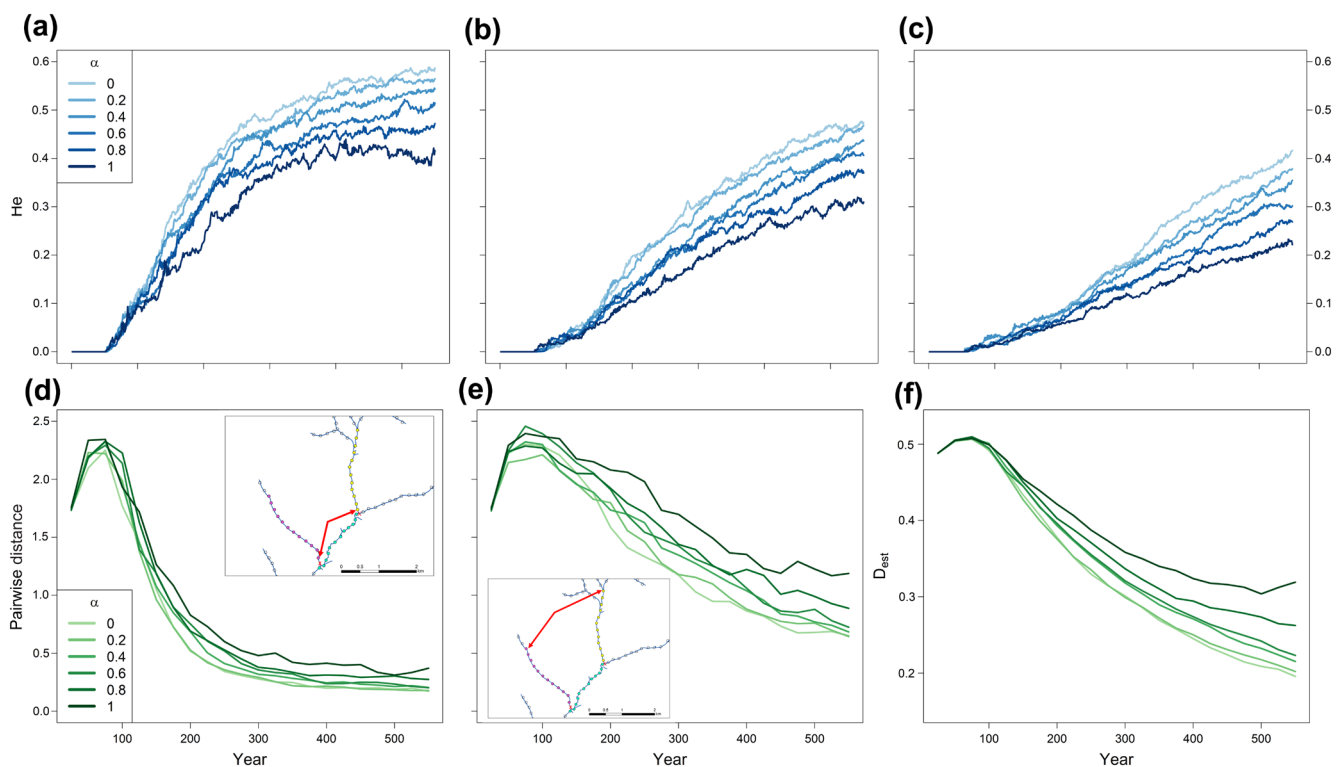


Figure 4. The top row of panels (a–c) presents the effects of interspecific competition on expected heterozygosity of the native westslope cutthroat trout population in the lower tributary. This population begins the simulation with extremely low genetic diversity and any increase in heterozygosity is attributed to gene flow from the upper tributary. From left to right, the line graphs display changes in expected heterozygosity over time as the observed patch moves from proximal (a) to distal (c) relative to the mainstem reach. The bottom row presents the effects of competition on population genetic structure, with panels (d) and (e) displaying pairwise genetic distance between the uppermost and lowermost patches of the two tributaries, respectively, as indicated by the red arrows. Panel (f) presents Jost's measure of global genetic structure for all patches over time.

spatially complex landscapes. The option to vary environmental attributes and interspecific competition coefficients spatially allows for the strength of competition to vary with the environment (Chesson 2000). Such capability may be used, for example, to ask questions about how landscape connectivity is different for populations occupying the core of the species range as opposed to the fringe where they may be interacting with competitor species (Case and Taper 2000). Similarly, CDMetaPOP 2 offers the ability to implement adaptive alleles so that probability of survival operates as a function of the individual's environment. Varying adaptation by species and allowing adaptive alleles to spread over time can be used to explore the role of evolutionary potential in the context of species interactions or to explore community-level selection and adaptation (Johnson and Boerlijst 2002). Finally, landscapes in CDMetaPOP 2 can be temporally dynamic in addition to being spatially explicit. This functionality allows users to swap in new landscapes according to past/future climate or management scenarios that may impact landscape connectivity differently for different species and likewise affect species interactions.

Approaches to measuring and defining functional landscape connectivity and landscape resistance have varied widely (Zeller et al. 2012). Species interactions can complicate measures of landscape resistance, particularly when the abundance of competitors and predators is not constant. Our simulation model offers a novel way to measure landscape connectivity and its effects on population genetics while mechanistically accounting for interspecific interactions. Rather than impose rules of landscape resistance due to species interactions, users can simulate competitor populations and measure functional connectivity as it emerges from model processes (Day et al. 2020b). This mechanistic approach is more transferable among species and landscapes and is more robust to alternative landscape scenarios than models where landscape connectivity is static or imposed (Grimm and Berger 2016, Radchuk et al. 2019).

CDMetaPOP 2 provides a flexible platform suitable for a wide range of empirical studies. Potential applications include conservation planning for communities (e.g. corridor development, habitat restoration), species reintroductions and translocations, competition within species guilds (i.e. 3+ species), population viability as a function of competition, invasion biology, evolution of co-occurring species, and range expansion/contraction, all in the context of adaptation. CDMetaPOP 2 is also well-suited for theoretical studies of intra- and interspecific competition in metapopulations and effects on landscape connectivity and population genetics. Finally, the development of the competition module provides a template for future modules such as predator/prey interactions, multispecies disease transmission, and mutualistic interactions.

Multispecies connectivity is becoming staple language in conservation planning, and a tool is needed that can simulate landscape genetics processes while accounting for interactions within communities and across trophic levels (Kissling and Schleuning 2015). Factors such as dispersal capacity and

suitability of dispersal habitat can vary widely among species in a community (Mims et al. 2018), and the optimal management strategy for one species may be implemented to the detriment of other species. If conservation efforts are going to be successful in stemming the global loss of biodiversity due to factors like land use change and climate change, conservation planning must consider species interactions and community-wide variation in genetic and species diversity (Mutshinda et al. 2011, Thomassen et al. 2011, Kissling and Schleuning 2015).

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Author contributions

Casey C. Day: Conceptualization (equal); Formal analysis (lead); Funding acquisition (equal); Investigation (equal); Methodology (lead); Software (supporting); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review and editing (lead). **Erin L. Landguth:** Conceptualization (equal); Data curation (supporting); Formal analysis (supporting); Funding acquisition (lead); Investigation (supporting); Methodology (supporting); Project administration (equal); Resources (equal); Software (lead); Supervision (equal); Validation (supporting); Visualization (supporting); Writing – original draft (supporting); Writing – review and editing (equal). **Ryan K. Simmons:** Conceptualization (equal); Data curation (lead); Formal analysis (supporting); Funding acquisition (lead); Investigation (supporting); Methodology (supporting); Project administration (lead); Resources (lead); Supervision (equal); Validation (supporting); Visualization (supporting); Writing – original draft (supporting); Writing – review and editing (equal). **Andrew R. Bearlin:** Conceptualization (equal); Data curation (equal); Funding acquisition (supporting); Resources (supporting); Software (supporting); Writing – review and editing (supporting).

Transparent peer review

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Data availability statement

Data are available from Github Repository: <https://github.com/ComputationalEcologyLab/CDMetaPop-v2-Ecography-Day-et-al.-2023> (Day et al. 2023).

Supporting information

The Supporting information associated with this article is available with the online version.

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